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Roll No: 2501730458
Subject: Nand2Tetris
Project: 3 - Sequential Chips
Section: G

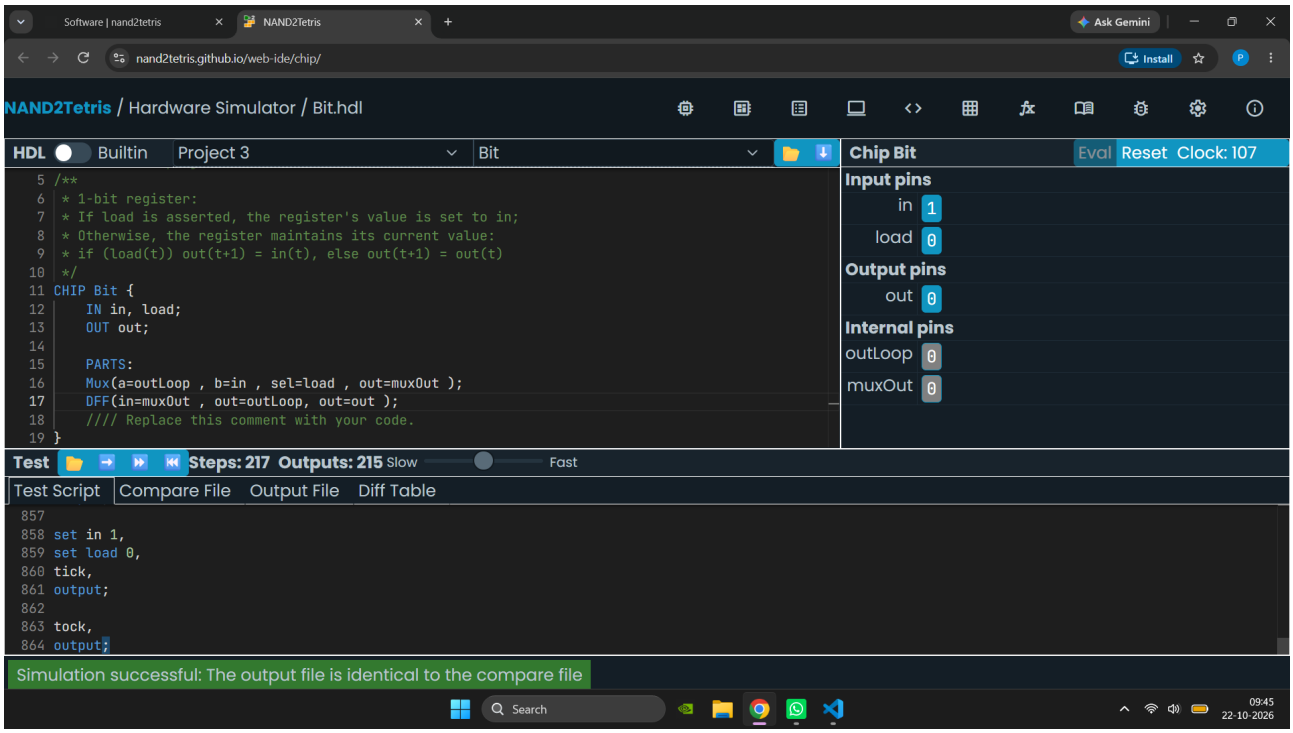
1. Bit.hdl (Project 3)

Code:

```
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/3/a/Bit.hdl
/**
 * 1-bit register:
 * If load is asserted, the register's value is set to in;
 * Otherwise, the register maintains its current value:
 * if (load(t)) out(t+1) = in(t), else out(t+1) = out(t)
 */
CHIP Bit {
    IN in, load;
    OUT out;

    PARTS:
    Mux(a=outLoop , b=in , sel=load , out=muxOut );
    DFF(in=muxOut , out=outLoop, out=out );
    //// Replace this comment with your code.
}
```

Screenshot (Hardware Simulator output):



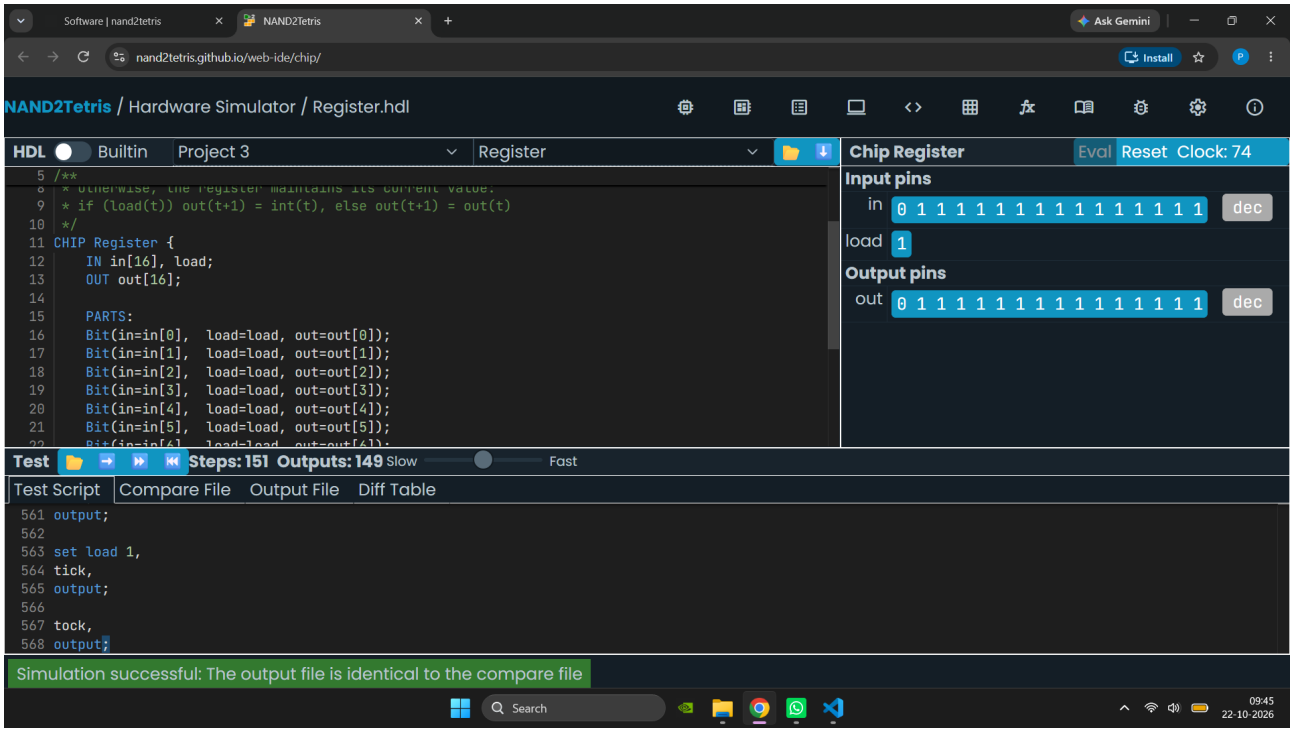
2. Register.hdl (Project 3)

Code:

```
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/3/a/Register.hdl
/**
 * 16-bit register:
 * If load is asserted, the register's value is set to in;
 * Otherwise, the register maintains its current value:
 * if (load(t)) out(t+1) = int(t), else out(t+1) = out(t)
 */
CHIP Register {
    IN in[16], load;
    OUT out[16];

    PARTS:
    Bit(in=in[0],  load=load, out=out[0]);
    Bit(in=in[1],  load=load, out=out[1]);
    Bit(in=in[2],  load=load, out=out[2]);
    Bit(in=in[3],  load=load, out=out[3]);
    Bit(in=in[4],  load=load, out=out[4]);
    Bit(in=in[5],  load=load, out=out[5]);
    Bit(in=in[6],  load=load, out=out[6]);
    Bit(in=in[7],  load=load, out=out[7]);
    Bit(in=in[8],  load=load, out=out[8]);
    Bit(in=in[9],  load=load, out=out[9]);
    Bit(in=in[10],  load=load, out=out[10]);
    Bit(in=in[11],  load=load, out=out[11]);
    Bit(in=in[12],  load=load, out=out[12]);
    Bit(in=in[13],  load=load, out=out[13]);
    Bit(in=in[14],  load=load, out=out[14]);
    Bit(in=in[15],  load=load, out=out[15]);
}
```

Screenshot (Hardware Simulator output):

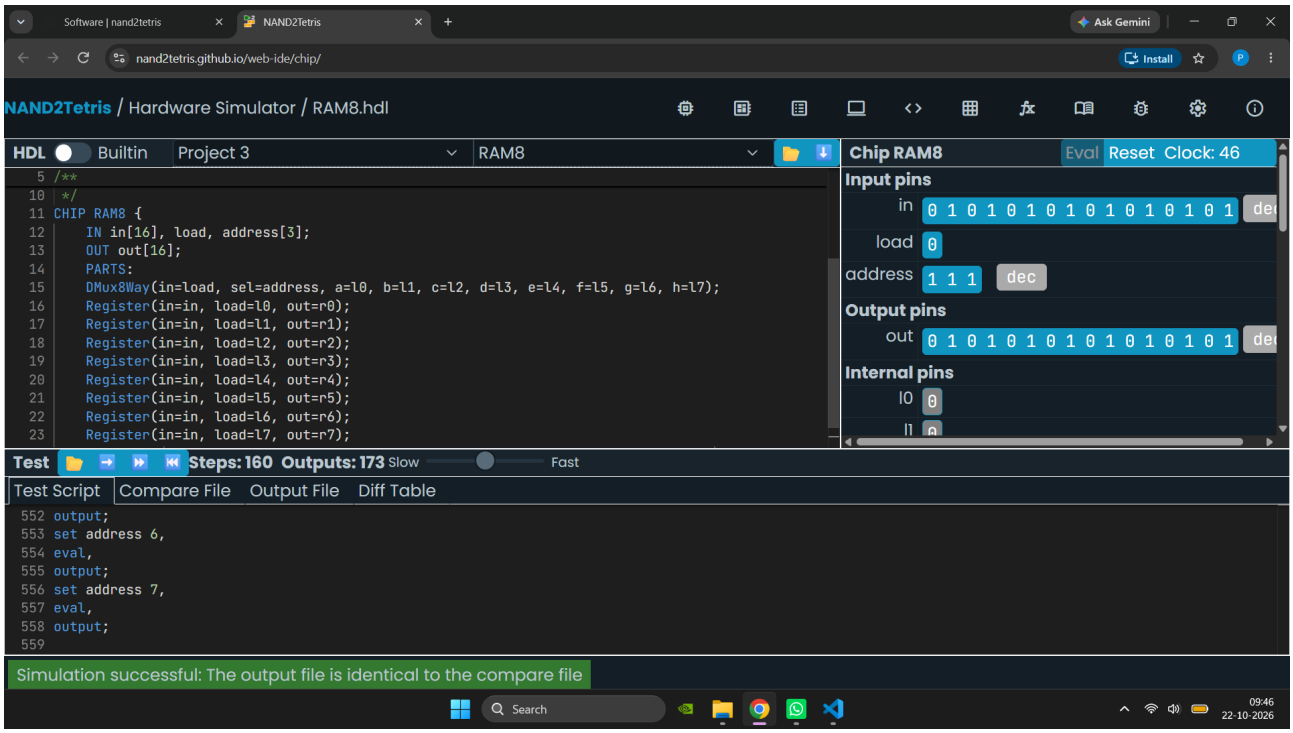


3. RAM8.hdl (Project 3)

Code:

```
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/3/a/RAM8.hdl
/**
 * Memory of eight 16-bit registers.
 * If load is asserted, the value of the register selected by
 * address is set to in; Otherwise, the value does not change.
 * The value of the selected register is emitted by out.
 */
CHIP RAM8 {
    IN in[16], load, address[3];
    OUT out[16];
    PARTS:
    DMux8Way(in=load, sel=address, a=l0, b=l1, c=l2, d=l3, e=l4, f=l5, g=l6, h=l7);
    Register(in=in, load=l0, out=r0);
    Register(in=in, load=l1, out=r1);
    Register(in=in, load=l2, out=r2);
    Register(in=in, load=l3, out=r3);
    Register(in=in, load=l4, out=r4);
    Register(in=in, load=l5, out=r5);
    Register(in=in, load=l6, out=r6);
    Register(in=in, load=l7, out=r7);
    Mux8Way16(a=r0, b=r1, c=r2, d=r3, e=r4, f=r5, g=r6, h=r7, sel=address, out=out);
}
```

Screenshot (Hardware Simulator output):

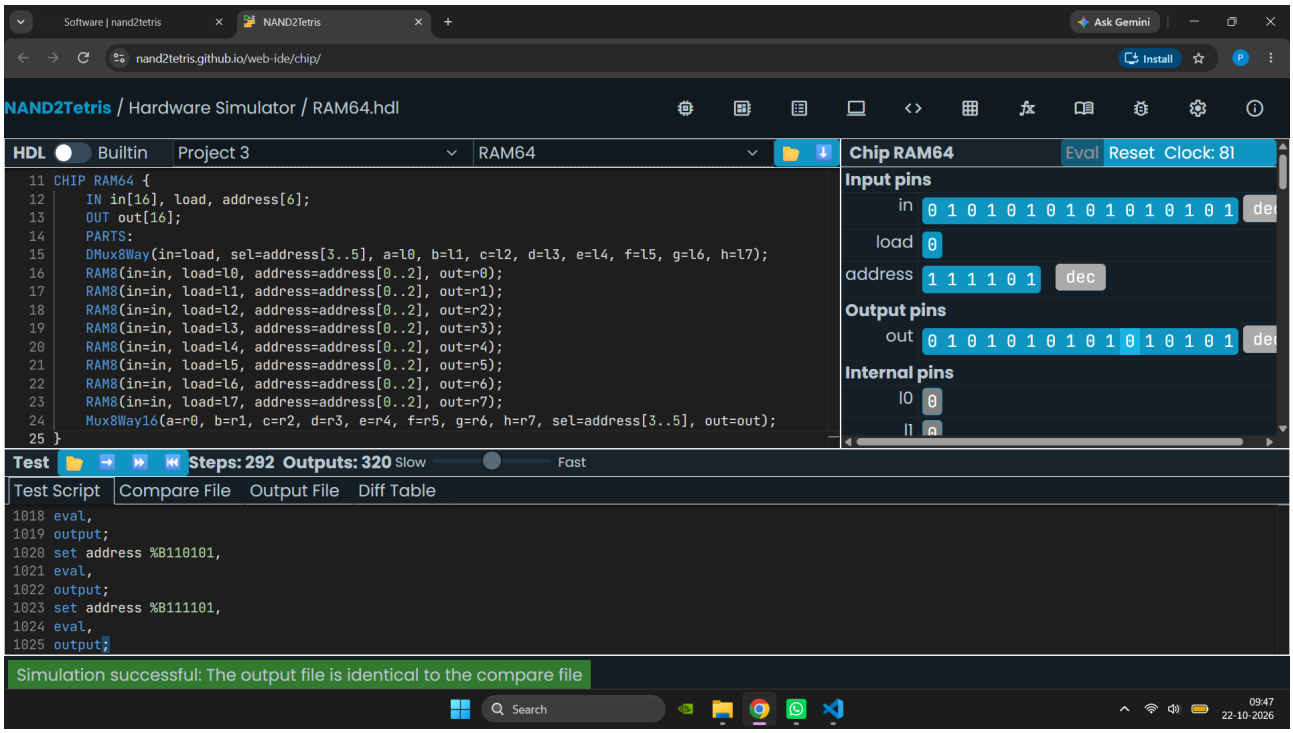


4. RAM64.hdl (Project 3)

Code:

```
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/3/a/RAM64.hdl
/**
 * Memory of sixty four 16-bit registers.
 * If load is asserted, the value of the register selected by
 * address is set to in; Otherwise, the value does not change.
 * The value of the selected register is emitted by out.
 */
CHIP RAM64 {
    IN in[16], load, address[6];
    OUT out[16];
    PARTS:
    DMux8Way(in=load, sel=address[3..5], a=l0, b=l1, c=l2, d=l3, e=l4, f=l5, g=l6, h=l7);
    RAM8(in=in, load=l0, address=address[0..2], out=r0);
    RAM8(in=in, load=l1, address=address[0..2], out=r1);
    RAM8(in=in, load=l2, address=address[0..2], out=r2);
    RAM8(in=in, load=l3, address=address[0..2], out=r3);
    RAM8(in=in, load=l4, address=address[0..2], out=r4);
    RAM8(in=in, load=l5, address=address[0..2], out=r5);
    RAM8(in=in, load=l6, address=address[0..2], out=r6);
    RAM8(in=in, load=l7, address=address[0..2], out=r7);
    Mux8Way16(a=r0, b=r1, c=r2, d=r3, e=r4, f=r5, g=r6, h=r7, sel=address[3..5], out=out);
}
```

Screenshot (Hardware Simulator output):

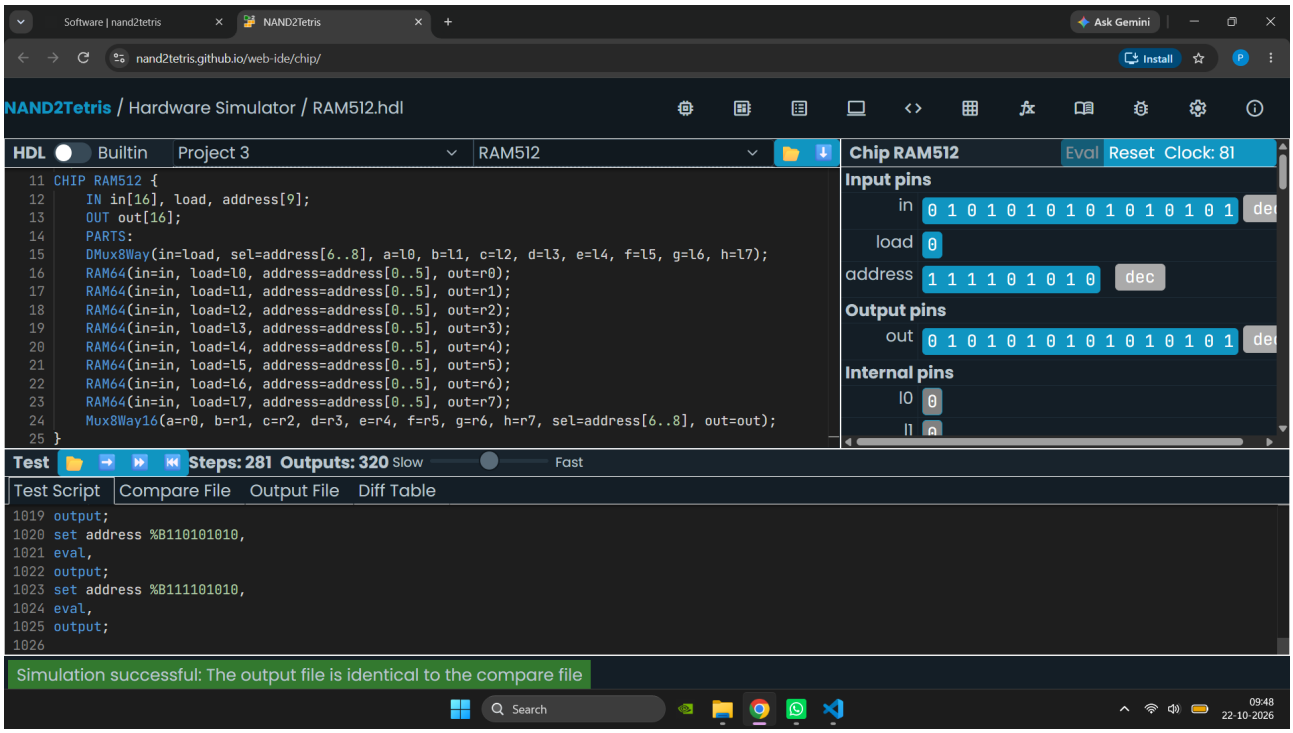


5. RAM512.hdl (Project 3)

Code:

```
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/3/b/RAM512.hdl
/**
 * Memory of 512 16-bit registers.
 * If load is asserted, the value of the register selected by
 * address is set to in; Otherwise, the value does not change.
 * The value of the selected register is emitted by out.
 */
CHIP RAM512 {
    IN in[16], load, address[9];
    OUT out[16];
    PARTS:
    DMux8Way(in=load, sel=address[6..8], a=l0, b=l1, c=l2, d=l3, e=l4, f=l5, g=l6, h=l7);
    RAM64(in=in, load=l0, address=address[0..5], out=r0);
    RAM64(in=in, load=l1, address=address[0..5], out=r1);
    RAM64(in=in, load=l2, address=address[0..5], out=r2);
    RAM64(in=in, load=l3, address=address[0..5], out=r3);
    RAM64(in=in, load=l4, address=address[0..5], out=r4);
    RAM64(in=in, load=l5, address=address[0..5], out=r5);
    RAM64(in=in, load=l6, address=address[0..5], out=r6);
    RAM64(in=in, load=l7, address=address[0..5], out=r7);
    Mux8Way16(a=r0, b=r1, c=r2, d=r3, e=r4, f=r5, g=r6, h=r7, sel=address[6..8], out=out);
}
```

Screenshot (Hardware Simulator output):

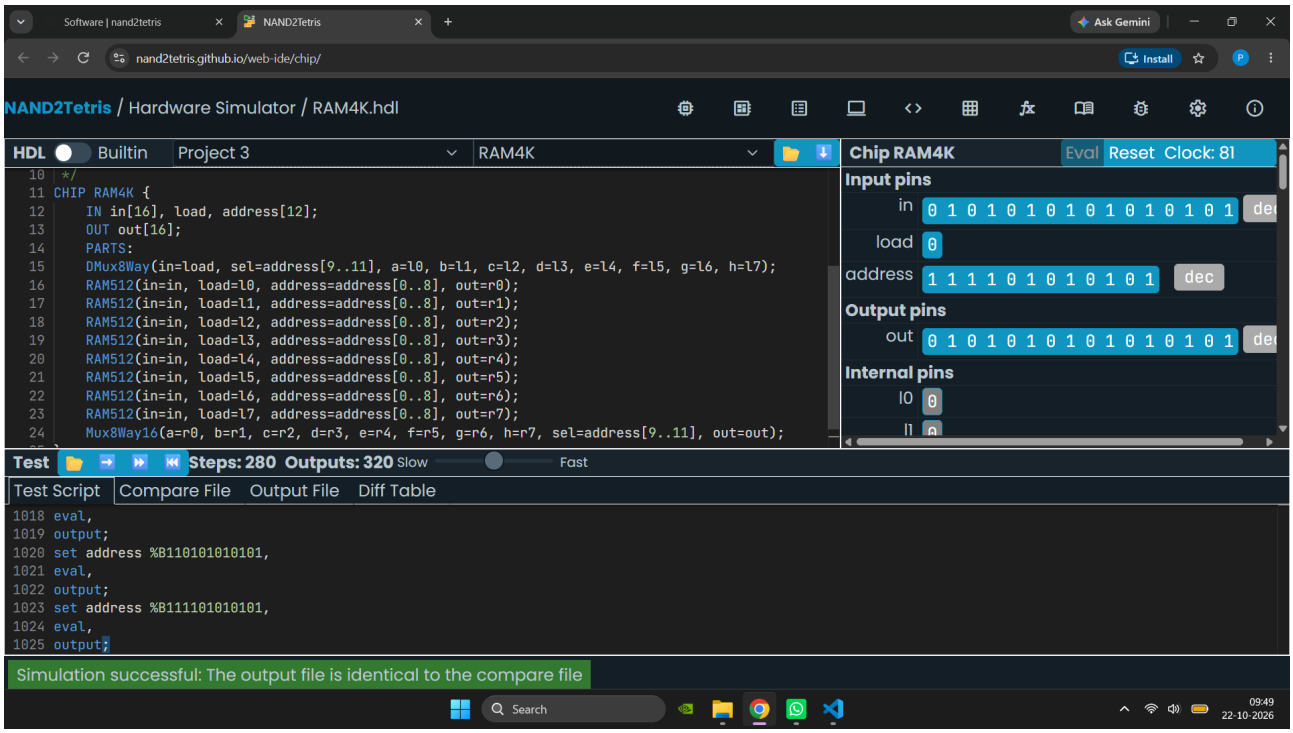


6. RAM4K.hdl (Project 3)

Code:

```
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/3/b/RAM4K.hdl
/**
 * Memory of 4K 16-bit registers.
 * If load is asserted, the value of the register selected by
 * address is set to in; Otherwise, the value does not change.
 * The value of the selected register is emitted by out.
 */
CHIP RAM4K {
    IN in[16], load, address[12];
    OUT out[16];
    PARTS:
    DMux8Way(in=load, sel=address[9..11], a=l0, b=l1, c=l2, d=l3, e=l4, f=l5, g=l6, h=l7);
    RAM512(in=in, load=l0, address=address[0..8], out=r0);
    RAM512(in=in, load=l1, address=address[0..8], out=r1);
    RAM512(in=in, load=l2, address=address[0..8], out=r2);
    RAM512(in=in, load=l3, address=address[0..8], out=r3);
    RAM512(in=in, load=l4, address=address[0..8], out=r4);
    RAM512(in=in, load=l5, address=address[0..8], out=r5);
    RAM512(in=in, load=l6, address=address[0..8], out=r6);
    RAM512(in=in, load=l7, address=address[0..8], out=r7);
    Mux8Way16(a=r0, b=r1, c=r2, d=r3, e=r4, f=r5, g=r6, h=r7, sel=address[9..11], out=out);
}
```

Screenshot (Hardware Simulator output):

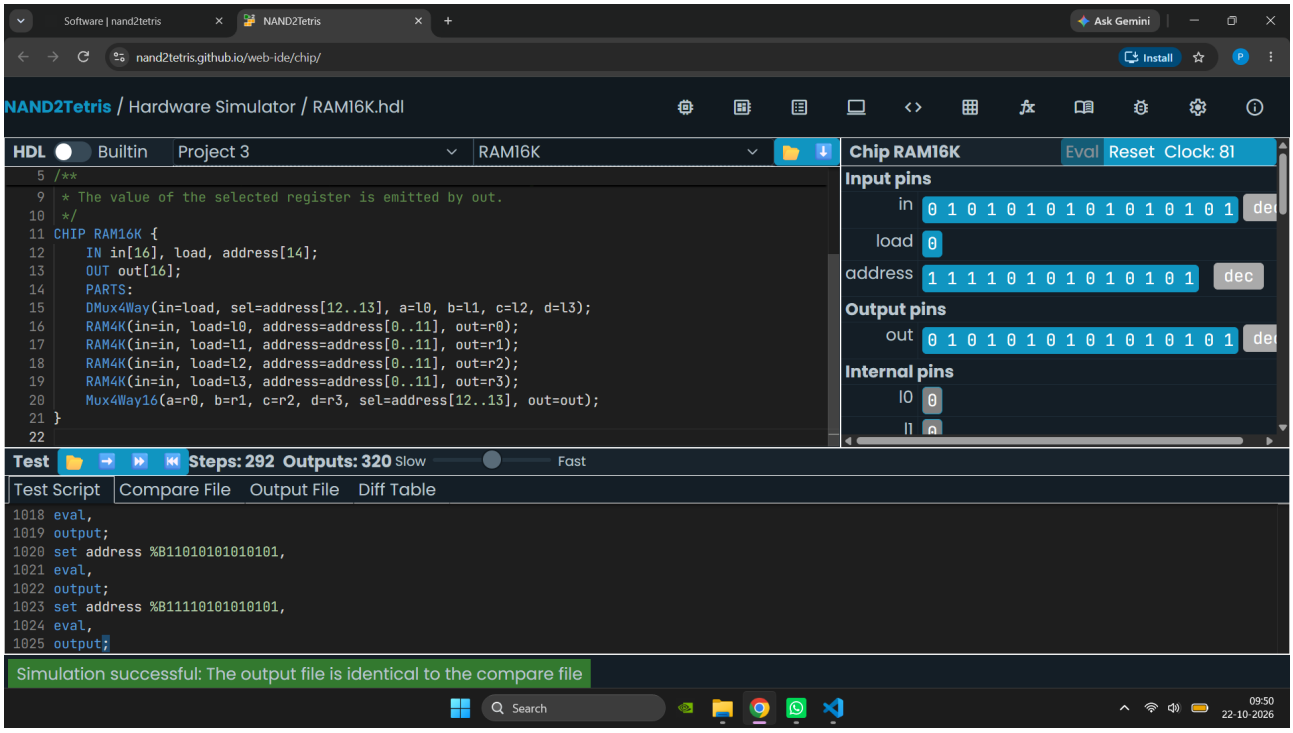


7. RAM16K.hdl (Project 3)

Code:

```
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/3/b/RAM16K.hdl
/**
 * Memory of 16K 16-bit registers.
 * If load is asserted, the value of the register selected by
 * address is set to in; Otherwise, the value does not change.
 * The value of the selected register is emitted by out.
 */
CHIP RAM16K {
    IN in[16], load, address[14];
    OUT out[16];
    PARTS:
    DMux4Way(in=load, sel=address[12..13], a=l0, b=l1, c=l2, d=l3);
    RAM4K(in=in, load=l0, address=address[0..11], out=r0);
    RAM4K(in=in, load=l1, address=address[0..11], out=r1);
    RAM4K(in=in, load=l2, address=address[0..11], out=r2);
    RAM4K(in=in, load=l3, address=address[0..11], out=r3);
    Mux4Way16(a=r0, b=r1, c=r2, d=r3, sel=address[12..13], out=out);
}
```

Screenshot (Hardware Simulator output):



8. PC.hdl (Project 3)

Code:

```
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/3/a/PC.hdl
/**
 * A 16-bit counter.
 * if      reset(t): out(t+1) = 0
 * else if load(t):  out(t+1) = in(t)
 * else if inc(t):   out(t+1) = out(t) + 1
 * else              out(t+1) = out(t)
 */
CHIP PC {
    IN in[16], reset, load, inc;
    OUT out[16];

    PARTS:
    Inc16(in=feedback, out=incOut);
    Mux16(a=feedback, b=incOut, sel=inc, out=incValue);
    Mux16(a=incValue, b=in, sel=load, out=loadValue);
    Mux16(a=loadValue, b=false, sel=reset, out=nextValue);
    Register(in=nextValue, load=true, out=out, out=feedback);
}
```

Screenshot (Hardware Simulator output):

